



Technical Service Bulletin: TSB150106	Released Date: 02-Sep-2015
Lubricating Oil Pan Adapter Gasket Repair Recommendations	

Lubricating Oil Pan Adapter Gasket Repair Recommendations

Core Issue

Customer may notice a lubricating oil leak from front and/or rear of engine at the lubricating oil pan adapter to cylinder block gasket as well as along the block rails at the lubricating oil pan adapter to cylinder block gasket.

The lubricating oil pan adapter gasket has **not** physically changed. This document informs the field of specific installation details and recommendations to be followed when repairing a lubricating oil leak from the lubricating oil pan adapter gasket. The procedure recommendations are being provided to help prevent repeat lubricating oil leaks.

Causes of the problem include.

- Insufficient crush on lubricating oil pan adapter rear rectangular seals.
- Front gear cover spacer plate misalignment.
- Lack of thread seals around lubricating oil pan adapter bolts.
- Insufficient bolt torque.

Confirmation

Product Affected

- K19 PT™ Fixed Time
- K19 HVT – Hydraulic Variable Timing
- K19 STC – Step Timing Control
- K19 CENTRY™ Fuel System
- K19G
- KTA19GC CM558/CM2358 SLB
- KTA19GC Rich Burn CM558
- G19
- QSK19 CM500 HPI®
- QSK19 CM850 MCRS
- QSK19 CM2150 MCRS
- QSK19G CM558 and CM700
- QSK19 CM2350 MCRS K105

To confirm lubricating oil is leaking from the lubricating oil pan adapter to cylinder block gasket, perform a Fluorescent Tracer Dye Test. Follow procedure 007-024 Lubricating Oil Leaks in the corresponding Service Manual, to verify the exact leak location is identified.

If the engine is under warranty, the exact leak location **must** be identified and described in the warranty claim.

Resolution

If the lubricating oil pan adapter to cylinder block gasket is identified as the source of the lubricating oil leak, Cummins Inc. recommends that the following steps be implemented while following repair instructions in procedure 007-027 Lubricating Oil Pan Adapter to help prevent reoccurring lubricating oil leaks.

- K19 or QSK19 engines with a paper gasket, Part Number 4000595, repair using a metal edge molded gasket, Part Number 2888747. See TSB120287.
- For QSK19 engines with an edge molded gasket, Part Number 2888747, repair with a like part (edge molded gasket).

Recommended RTV to be used in instructions below:

- Loctite® 5699 - Grey RTV 190-ml Power Can, Cummins® service tool part number 4918882.
- Loctite® 5999 - Grey RTV 300-ml Cartridge, Cummins® service tool part number 3164070.

In addition to the installation instructions in procedure 007-027 Lubricating Oil Pan Adapter, pay special attention to the following instructions when repairing with the metal edge molded gasket.

- Pay close attention to the front T-joints. The front gear cover, front gear cover gasket, gear cover spacer plate and gear cover spacer plate gasket (Figure 3, Callout 5) **must** be flush with the bottom of the cylinder block to promote lubricating oil pan gasket sealing.
 - If the gear cover spacer plate is removed, adjust the diamond dowel in the block so the gear cover spacer plate does **not** hang below the block surface on both sides.
 - Diamond Dowel (Figure 1, Callout 1) – Turning this dowel adjusts the gear cover spacer plate. Adjust so gear cover spacer plate sits flush with the bottom of the cylinder block at the front corners (T-joint) on each side of the cylinder block.
 - Master Dowel (Figure 1, Callout 2) – Flat on diamond dowel (1) typically will face the master dowel (2) for optimal spacer plate alignment. This will **not** be accurate on all engines. Adjust the diamond dowel (1) for the best alignment.
 - Bolt a straight edge (Figure 2, Callout 3) under the gear cover spacer plate utilizing a cylinder block bolt hole (Figure 2, Callout 4) on both sides to confirm the gear cover spacer plate stays flush with the bottom of the cylinder block while the gear cover spacer plate is installed and torqued down. See Figures 1-3.



Figure 1

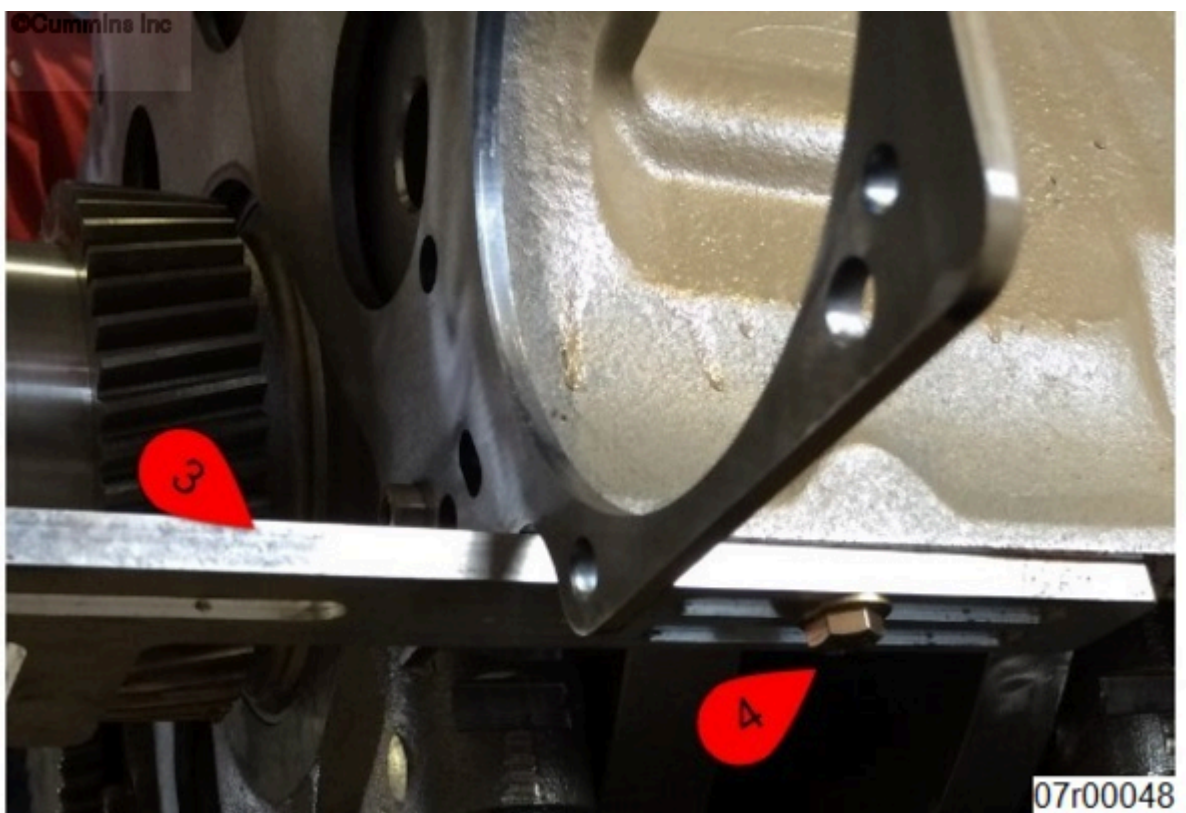


Figure 2

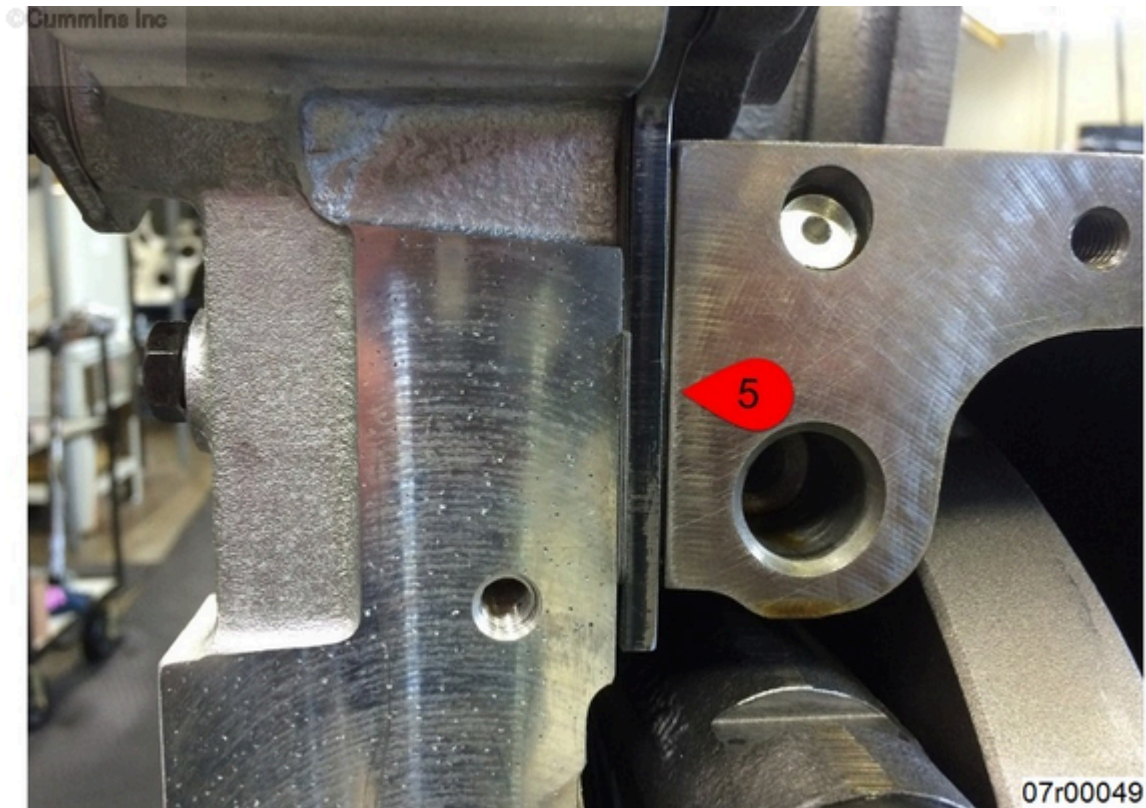


Figure 3

- Verify cylinder block and lubricating oil pan adapter gasket surfaces and any joints are clean and dry to promote optimal gasket sealing and RTV adhesion.
 - This includes wiping down the new lubricating oil pan adapter gasket and new rear lubricating oil pan adapter square seals before applying RTV to them.
- Apply sealant to front T-joint gaps on both sides of cylinder block and push the sealant up into the gasket void with your finger where the gear cover spacer plate gasket resides (Figure 4). Use Cummins® service tool part numbers listed above, Loctite® 5699 or 5999. This step fills the void with sealant, eliminating a potential lubricating oil leak path.



Figure 4

- Using a clean cloth, wipe off any excess sealant (Figure 5, Callout 1) and lay a bead over the joint (Figure 6, Callout 1) as stated in procedure 007-027 Lubricating Oil Pan Adapter.

Note : The steps above are especially important on QSK19 engines that have the metal edge molded gear cover spacer plate gasket as this gasket does not extend down to the bottom of the cylinder block. This is normal and by design. All K19 and some QSK19 engines before Engine Serial Number (ESN) 37254361 will have the paper gear cover spacer plate to cylinder block gasket. Verify the paper gasket is flush with the bottom of the cylinder block and does not protrude in any way.

⚠ CAUTION ⚠

The metal edge molded gear cover spacer plate gasket, Part Number 4922528, is used exclusively on QSK19 engines and is not compatible with K19 engines. Only use part number 4922528 on QSK19 engines.



Figure 5

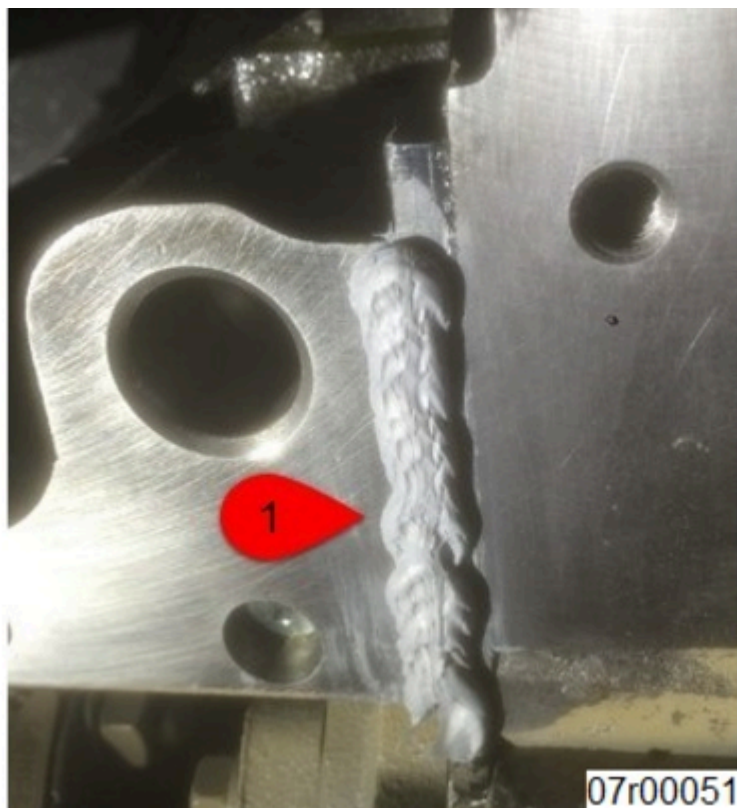


Figure 6

- When applying sealant on the cylinder block at the rear corners in front of flywheel housing; apply sealant in L pattern so gap between edge of lubricating oil pan adapter gasket and rectangular seal is filled when the lubricating oil pan adapter is installed. Use Cummins® service tool part numbers listed above, Loctite® 5699 or 5999.
 - Sealant applied on cylinder block at right rear (Figure 7, Callout 1).
 - Sealant applied on cylinder block at left rear (Figure 7, Callout 2).

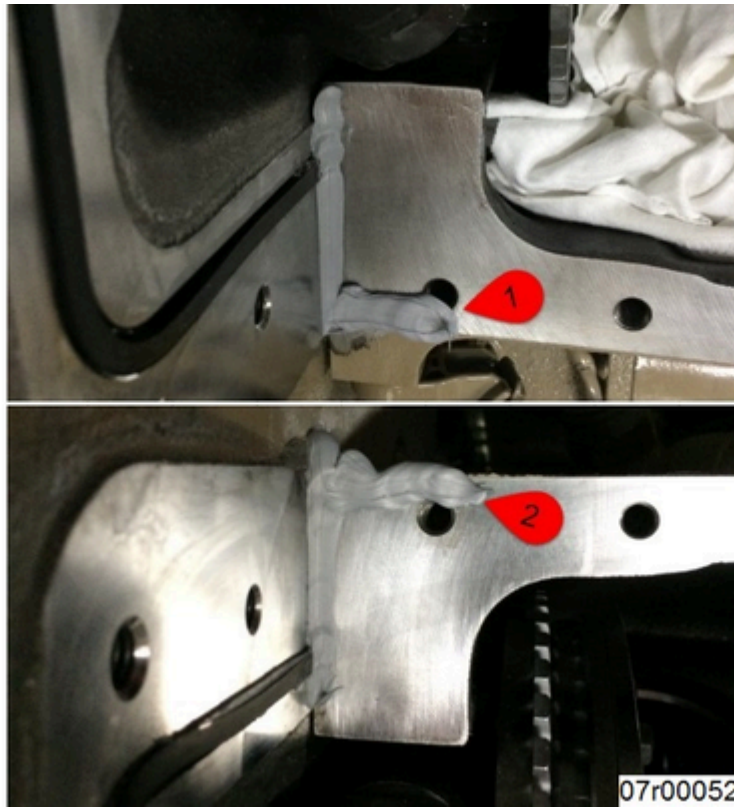


Figure 7

- Apply a 3mm [1/8 in] thick bead of sealant on **both** sides of the lubricating oil pan adapter gasket along the outside of the bolt holes (Yellow Line in Figure 8).
 - Be sure to connect the bead of sealant with the front and rear T-joint sealant on the cylinder block to eliminate a potential lubricating oil leak path.

⚠ CAUTION ⚠

RTV must be kept out of and away from the lubricating oil suction tube hole in the cylinder block or engine damage will result. (Figure 8, Callout 2).

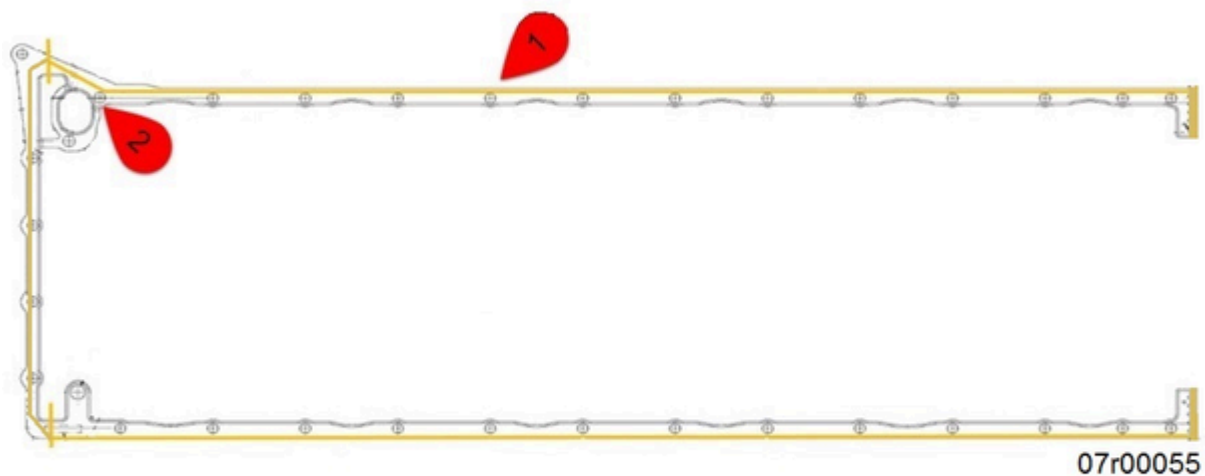


Figure 8

- Tighten the lubricating oil pan adapter bolts using the correct tightening sequence and torque value. Tighten the bolts a 2nd time to the final torque value using the correct tightening sequence. See procedure 007-027 Lubricating Oil Pan Adapter in the corresponding Service Manual.

Service Parts Availability

Service parts are available. See Table 1 for part numbers.

Table 1, Lubricating Oil Pan Adapter Gasket Part Numbers	
Part Description	Part Number
Gasket, Oil Pan Adapter (Paper)	4000595
Gasket, Oil Pan Adapter (Metal Edge Molded)	2888747

Production Status

Addition of Loctite™ 5999 RTV along outside of capscrew holes (see Figure 8 above) on both sides of lubricating oil pan adapter gasket was implemented in production for QSK19 engines **only**. See Table 2 below.

Note : RTV application to both sides of the lubricating oil pan adapter gasket is considered a temporary countermeasure until a more permanent solution is implemented.

Table 2, Production Information – RTV Applied to Lubricating Oil Pan Adapter Edge Mold Gasket – QSK19 Engines Only		
Engine Serial Number (ESN) First	Build Date ¹	Plant
37266001	21-July-2015	Seymour Engine Plant

Table 2, Production Information – RTV Applied to Lubricating Oil Pan Adapter Edge Mold Gasket – QSK19 Engines Only		
Engine Serial Number (ESN) First	Build Date¹	Plant
¹ Engine build date can be found on the engine dataplate.		

Associated Publications

- K19 Service Manual, Bulletin Number 4021499
- QSK19, QSK19 CM850 MCRS, and QSK19 CM2150 MCRS Service Manual, Bulletin Number 4021592
- QSK19 CM2350 K105 Service Manual, Bulletin Number 4326157
- KTA19GC CM558 Service Manual, Bulletin Number 4021653
- QSK19G Generator Set Troubleshooting and Repair Manual, Bulletin Number 4021372

Warranty Statement

The information in this document authorizes specific changes to the repair practice for failures covered under product warranty coverages.

Document History

Date	Details
2015-7-6	Module Created
2015-7-24	Updated production status and Table 1.
2015-8-31	Product Problem Solving (PPS) Step 3 Action

Last Modified: 02-Sep-2015
